Ariel
RENEWABLE ENERGY LIMITED

SOVEREIGN GREEN INFRASTRUCTURE & INDUSTRIAL DECARBONIZATION



TURNKEY SOLAR EPC & SREDA NET-METERING COMMERCIAL DOSSIER

A Strategic Capital Investment, Technical Architecture & 25-Year Decarbonization Framework for Bangladesh Industrial, Commercial & Institutional Facilities

DOCUMENT CLASSIFICATION: Strictly Private & Confidential	**ISSUING ENTITY:** Ariel Renewable Energy Limited
DOCUMENT REFERENCE: AREL-EPC-2026-BOARD-V1	**PARENT GROUP:** ArielMC Enterprise Eco-System
PREPARED FOR: The Board of Directors / Chairman / CEO	**SECTOR / PRODUCT:** Energy / Solar Infrastructure
REGULATORY FRAMEWORK: SREDA Net-Metering Guidelines 2018	**FINANCING SCHEME:** Bangladesh Bank 4.5% Green Refinancing
COMMAND HUB: energy.arielmc.com | crm.arielmc.com	**DATE OF ISSUANCE:** November 2026 (Commercial Release)

Notice of Confidentiality: This commercial and technical dossier contains proprietary engineering architectures, financial models, and strategic methodologies of Ariel Renewable Energy Limited. It is prepared solely for the evaluated client board and shall not be reproduced or distributed without prior written consent.

EXECUTIVE TRANSMITTAL NOTE

Official Address from the Managing Director to the Client Board of Directors

To: The Chairman, Managing Director & Members of the Board
Subject: Strategic Turnkey Rooftop Solar EPC & Long-Term Energy Cost Insulation

Dear Industrial & Commercial Leaders,

As Bangladesh navigates rapid industrial transformation, energy cost and grid reliability have become the single most decisive determinants of export competitiveness and enterprise profitability. Over the past 36 months, bulk industrial electricity tariffs have escalated continuously, while fossil-fuel supply shocks and captive diesel generation costs (exceeding Tk 32 to 36 per kWh) place unprecedented strain on operational margins.

Concurrently, European and American buyers are strictly enforcing decarbonization mandates. The European Union's **Carbon Border Adjustment Mechanism (CBAM)**, LEED Platinum factory standards, and global Scope-2 greenhouse gas auditing mean that clean power is no longer an optional corporate luxury—it is an existential commercial requirement to protect export volume.

Your rooftop represents an unharvested goldmine. A vacant industrial PEB shed, commercial tower slab, or institutional roof can immediately be converted into a high-yielding, captive power plant producing clean daylight electricity at an amortized Levelized Cost of Electricity (LCOE) of **Tk 3.20 per unit**—compared to the Tk 11.50 to Tk 15.00 per unit you currently pay the national grid.

Ariel Renewable Energy Limited, a proud subsidiary of the **ArielMC** enterprise group, provides a total, zero-risk turnkey solution. We assume full legal, structural, civil, and regulatory responsibility: from 3D LiDAR drone surveying and SREDA Net-Metering approvals to high-spec EPC installation and 25 years of guaranteed Operations & Maintenance.

This dossier lays out our engineered packages, transparent Tier-1 Bill of Quantities, bankable financial returns, and risk-elimination frameworks. We welcome the opportunity to present these findings to your board and conduct a complimentary, non-intrusive drone roof feasibility audit at your earliest convenience.

Sincerely and with national pride,

Syed Akramul Hassan
Managing Director & Chief Executive Officer
Ariel Renewable Energy Limited | *ArielMC Enterprise Eco-System*
Dhaka, Bangladesh • energy.arielmc.com • crm.arielmc.com



Figure 1: Ariel Renewable Energy certified solar electrical and structural engineering division conducting precision digital auditing and telemetry commissioning.

1. THE BANGLADESH ENERGY PARADIGM & ECONOMIC IMPERATIVE

Escalating Grid Tariffs, Decarbonization Mandates & The SREDA Net-Metering Engine

Industrialists and facility owners in Bangladesh currently face a converging triple energy challenge: historical grid tariff inflation averaging 12–15% annually, constrained natural gas supply forcing reliance on expensive captive liquid fuels, and international buyer ESG scrutiny. The table below illustrates the stark economic contrast between legacy grid power and on-site solar generation:

Energy Source / Tariff Class	Effective Cost / kWh (BDT)	Cost Trend (2026–2035)	Carbon / Buyer Audit Status	Grid / Supply Reliability
REB / DESCO Industrial (LT-E / HT-3)	Tk 10.80 – 12.20 / unit	Escalating +8% to +12%/yr	Fossil-heavy; Scope 2 penalty	Subject to seasonal load curtailment
Commercial Peak Grid (Motijheel/Gulshan)	Tk 13.80 – 15.20 / unit	Escalating +10% to +14%/yr	Zero green points; high OPEX	High peak-hour demand penalties
Captive Diesel Generator (Backup)	Tk 32.00 – 38.00 / unit	Tied to volatile crude imports	Severe carbon/smoke emissions	Emergency only; mechanical wear
Ariel Turnkey Rooftop Solar (SREDA)	**Tk 3.20 / unit (LCOE)**	**FIXED for 25 Years (Tk 0.00 Fuel)**	**100% Green; CBAM & LEED Certified**	**Daylight Base-Load Guarantee**

The Mechanics of the SREDA Net-Metering Advantage:

Under the Government of Bangladesh's *Net-Metering Guidelines 2018 (and gazetted revisions)*, any commercial or industrial consumer can connect an approved rooftop solar system directly to the utility grid via a bi-directional (import/export) smart meter. During peak daylight hours (9:00 AM – 3:30 PM), the solar array first feeds the facility's internal operating machines. Any surplus electricity automatically flows into the national grid. At the end of the monthly billing cycle, utility companies (REB, DESCO, DPDC, BPDB) credit every exported kilowatt-hour against the facility's evening or weekend consumption at the full published tariff rate. If exports exceed imports, the monetary credit rolls over indefinitely to offset future bills.



Figure 2: Ariel Renewable Energy 6.1 MWp Multi-Shed Continuous Rooftop Solar Array energizing a LEED Platinum Industrial Complex.

2. ENGINEERED COMMERCIAL PACKAGES — INDUSTRIAL SECTOR

Heavy Industrial, Export RMG, Textiles, Leather Tannery & Jute Mills

Ariel Renewable Energy has engineered three distinct modular tiers for industrial operations. Each package is tailored to facility electrical demand, roof profile (PEB Corrugated Metal, Standing Seam, or RCC Slab), and substation connection topology.

Package Code & Title	System Capacity	Required Roof Space	Estimated Monthly Yield	Grid Interconnection	Target Facility Profile
F1: Industrial Base-Load Offset	**300 kWp** (512 Modules)	~24,000 sq ft Corrugated / PEB	**36,000 – 39,000 kWh** (Monthly)	400V LT / 11kV HT Net-Metering	Washing, dyeing, leather finishing, medium manufacturing sheds.
F2: Megawatt Flagship *(Most Popular)*	**1,000 kWp (1.0 MWp)** (1,710 Modules)	~78,000 sq ft Single / Dual Shed	**125,000 – 135,000 kWh** (Monthly)	11kV / 33kV Dedicated Substation Sync	Composite knit/woven mills, spinning mills, export footwear, ceramics.
F3: Net-Zero Multi-Roof Complex	**2,500 kWp (2.5 MWp)** (4,275 Modules)	~200,000+ sq ft Multi-Shed Campus	**315,000 – 340,000 kWh** (Monthly)	33kV Express Feeder + Ariel SCADA Console	Large conglomerate industrial parks, heavy steel, EPZ export complexes.



Left: 3.8 MWp Riverine Export Textile Mill (Gazipur) | Right: 5.2 MWp Mega-Scale Standing-Seam PEB Apparel Complex (Ashulia).

3. COMMERCIAL BUILDINGS & HIGH-END RESIDENTIAL

Corporate Headquarters, Private Universities, Hospitals & Apartment Communities

Category 2: Commercial Buildings & Institutional Real Estate

Commercial facilities carry the highest published electricity tariffs in Bangladesh (Tk 13.80 – 15.20/kWh) with day-time air conditioning driving intense peak demand. Solar arrays harvest maximum power during precisely these hours, slashing peak demand penalties with rapid 3-year payback.

Package Code & Title	Capacity	Roof Footprint	Monthly Generation	Primary Application
C1: Corporate Tower Rooftop	**75 kWp**	~6,000 sq ft RCC	**9,000 – 10,000 kWh**	10–15 storey corporate office towers in Gulshan, Banani, Motijheel, Agrabad.
C2: Retail Mall & Hospital	**200 kWp**	~16,000 sq ft	**25,000 – 27,000 kWh**	Shopping complexes, multi-specialty private hospitals, hotel roof slabs.
C3: Campus Canopy & Carport	**500 kWp**	~40,000 sq ft	**62,000 – 68,000 kWh**	Private university campuses, international schools, convention centers, solar parking lots.

Category 3: Household & Residential Communities

Package Code & Title	Capacity	Space Req.	Monthly Yield	System Highlights
H1: Urban Villa Micro-Array	**10 kWp Grid-Tied**	~800 sq ft RCC	1,200 – 1,350 kWh	Slashes domestic top-tier slab rates (Tk 13.50/unit) for luxury villas & duplexes.
H2: Luxury Hybrid + LFP Storage	**20 kWp + 15 kWh**	~1,600 sq ft RCC	2,400 – 2,600 kWh	Zero generator noise; 10ms seamless battery switchover during grid blackouts.
H3: Apartment Society Common-Area	**40 kWp Comm-Grid**	~3,200 sq ft Roof	5,000 – 5,400 kWh	Powers common lifts, stair lighting & water pumps; slashes monthly service charge 40–55%.



Figure 3: High-Rise Commercial Corporate Tower (Gulshan/Tejgaon) equipped with elevated solar rooftop pergolas and smart telemetry.

4. TECHNICAL ARCHITECTURE & MULTI-OPTION HARDWARE

Tier-1 BloombergNEF PV Modules, Multi-MPPT Inverters, Heavy Structural Frames & IoT SCADA

Ariel Renewable Energy maintains an uncompromising hardware selection policy. Every system component is sourced strictly from Tier-1 manufacturers holding global IEC/UL certifications and verified field track records across tropical monsoon operating environments.

Component Architecture	Tier-1 Premium Option	Tier-1 Value-Optimized Option	Engineering Standards & Warranties
Photovoltaic Solar Modules (Bifacial Dual-Glass)	**Jinko Solar Tiger Neo 620W** • N-Type TOPCon Dual-Glass • Module Efficiency: 22.8% • Bifaciality Factor: 80±5% • Annual degradation: ≤ 0.40%/yr	**LONGi Hi-MO X6 / Trina 585W** • Monocrystalline N-Type • Module Efficiency: 21.8% • Anti-PID, anti-leTID protection • Salt-mist & ammonia resistant	• **12 Years** Product Workmanship • **30 Years** Linear Performance (≥87.4% output at Year 30) • IEC 61215, IEC 61730, UL 61730 certified
Grid-Tied String Inverters (Multi-MPPT High Efficiency)	**Huawei SUN2000 Series** • 50KTL / 100KTL / 115KTL-M2 • Max Efficiency: 98.8% • AI-Powered AFCI arc protection • Smart IV Curve Diagnosis 4.0	**Sungrow SG Series** • SG50CX / SG110CX Multi-MPPT • Max Efficiency: 98.6% • IP66 / C5 anti-corrosion rated • Dynamic PID recovery built-in	• **5 Years** Standard Manufacturer Warranty (Extendable to 10 Years) • VDE-AR-N 4105, EN 50549-1, IEEE 1547 compliant
Mounting Substructures (Non-Penetrative & Elevated)	**Aluminium 6005-T5 Anodized** • High-strength extruded alloy • Non-penetrative standing seam clamps • Corrosion Resistance: Marine C5-M • Zero roof drill puncture	**Hot-Dip Galvanized Steel (HDG)** • 80–100 micron galvanization • SS304 / SS316 marine fasteners • Ballasted & elevated RCC structures • Custom wind-tunnel tested frame	• **15 Years** Structural Warranty • **180 km/h** Wind Load Certified (BNBC 2020 Cyclone Code) • Zero roof leakage structural seal guarantee
Switchgear & Surge Protection (Class-A Industrial Protection)	**ABB / Schneider Electric** • DC Isolators 1000V/1500V rated • Type 1+2 DC/AC Surge Arresters • Copper busbars & motorized MCCBs • Shunt trip emergency cutoff	**Dehn / Chint High-Spec** • IP65 outdoor combiner enclosures • Dual-core Class 0.2S Net-Meter • Heavy copper tape down conductors • Deep earth boreholes (< 1 Ohm)	• Factory type-tested assemblies • SREDA Net-Metering compliant • True dry earthing isolation for sensitive manufacturing equipment
Telemetry & Cloud SCADA (Ariel-SaaS™ Command Suite)	**Huawei FusionSolar Smart PV MS** • String-level power tracking • Real-time cloud gateway via 4G/LAN • Executive iOS / Android app • Autonomous yield & fault alarms	**Sungrow iSolarCloud Console** • High-precision Modbus datalogger • Weather station & pyranometer sync • SREDA export credit ledger • Daily automated yield reports	• **25-Year** Cloud Monitoring License • Live sync with ArielMC Master-CRM & Ariel-SaaS Enterprise Command Console

5. TURNKEY INVESTMENT COSTING & 25-YEAR ROI

All-Inclusive Turnkey Pricing (BDT/USD), Detailed BOQ Breakdown & Bangladesh Bank 4.5% Debt Model

Ariel Renewable Energy delivers transparent, end-to-end EPC contract pricing. Our commercial bids include 100% of civil engineering, imported Tier-1 equipment, customs clearance, mechanical erection, electrical cabling, utility synchronization, and SREDA licensing.

Target Customer Package	Capacity (kWp)	Rate (Tk/Wp)	Rate ($/Wp)	Total Turnkey CapEx (BDT)	Total CapEx (USD)	Simple Payback
F1: Industrial Base-Load Offset	300 kWp	Tk 52.00	$0.433	**Tk 1,56,00,000**	$130,000	**3.2 Years**
F2: Megawatt Flagship	1,000 kWp (1.0 MWp)	Tk 49.00	$0.408	**Tk 4,90,00,000**	$408,300	**3.0 Years**
F3: Heavy Industrial Complex	2,500 kWp (2.5 MWp)	Tk 47.50	$0.395	**Tk 11,87,50,000**	$989,500	**2.85 Years**
C1: Corporate Office Tower	75 kWp	Tk 62.00	$0.516	**Tk 46,50,000**	$38,750	**3.1 Years**
C2: Retail Mall & Hospital	200 kWp	Tk 58.00	$0.483	**Tk 1,16,00,000**	$96,600	**2.9 Years**
H1: Urban Villa Micro-Array	10 kWp Grid-Tied	Tk 75.00	$0.625	**Tk 7,50,000**	$6,250	**4.2 Years**
H2: Luxury Hybrid + 15kWh LFP	20 kWp + Battery	Tk 110.00	$0.916	**Tk 22,00,000**	$18,300	**5.5 Years**
H3: Apartment Society Common	40 kWp Comm-Grid	Tk 68.00	$0.566	**Tk 27,20,000**	$22,600	**3.8 Years**

Detailed BOQ Capital Allocation & 80% Green Debt Structuring

For every Tk 100 invested: **43%** Tier-1 PV Modules, **18%** Multi-MPPT Inverters & Protection Switchgear, **12%** Anodized Aluminium/HDG Structural Frames, **8%** Solar DC/AC XLPE Cables & Conduit, **6%** Substation & Net-Meter Synchronization, **7%** Civil Work & Rigging, **6%** Engineering Design, SREDA Licensing & Testing.

Zero Equity Depletion via Bangladesh Bank Green Refinance Scheme: Under the Bangladesh Bank Green Transformation Fund (GTF) and SREDA Solar Refinance Window, eligible manufacturing enterprises can finance up to **80% of project CapEx at a 4.5% to 5.0% concessionary interest rate** over a 5 to 7 year repayment tenure. Because monthly grid electricity savings substantially exceed the monthly loan installment, the solar plant generates **positive net cashflow from the very first month of operation**.

6. STRUCTURAL SAFETY, CIVIL INTEGRITY & REGULATORY PATHWAY

Zero-Penetration Roof Waterproofing, BNBC 2020 Cyclone Code & Turnkey SREDA Approval

Industrial CEOs and plant directors often express legitimate concerns regarding structural roof integrity, monsoon leakage risks, and government bureaucracy. Ariel Renewable Energy has engineered robust protocols to eliminate these risks entirely:

Risk Concern	Ariel Renewable Energy Technical Defense Protocol	Legal / Certified Guarantee
Roof Water Leakage (Monsoon Rains)	For standing-seam PEB metal sheds, we employ **100% non-penetrative aluminium seam clamps**—zero sheet drilling. For trapezoidal sheet roofs, we utilize EPDM-gasketed self-tapping fasteners with Sikaflex UV sealant. On RCC slabs, ballasted mountings or double-chemical-anchored stanchions ensure zero water pooling.	**15-Year 100% Water-Tight Guarantee** backed by written EPC contract liability.
Cyclonic Wind Loads (Monsoon Storms)	Mounting structures are designed by certified structural engineers in accordance with **BNBC 2020 Wind Zone Standards** (tested to withstand continuous wind gusts of up to **180 km/h**). Aerodynamic wind deflectors reduce uplift pressure.	Certified Structural Engineer (BSc/MSc Civil BUET) load vetting stamp provided for every project.
Fire & Electrical Arcs (Industrial Safety)	Inverters feature embedded **AI-powered Arc Fault Circuit Interrupters (AFCI)** that detect and extinguish DC electrical arcs within 2.5 seconds. Halogen-free flame-retardant DC cables, IP65 combiner boxes, and isolation transformers eliminate fire hazards.	NFPA 855 and IEC 62109 fire safety compliance.
Lightning & Surges (Tropical Storms)	Dedicated copper Franklin lightning interception masts with independent down conductors and chemical earthing pits achieve resistance **< 1.0 Ohm**, completely isolating the array from sensitive manufacturing machinery.	IEC 62305 lightning protection certification.

Turnkey 5-Stage SREDA Regulatory & Grid Interconnection Roadmap

Ariel Renewable Energy handles 100% of the administrative and technical regulatory process:

1. Stage 1: Site Feasibility & 3D Drone Shadow Modelling (Days 1–5): High-resolution aerial LiDAR mapping, structural load testing, and PVsyst yield forecasting.

2. Stage 2: Utility Load Sanction & Substation Verification (Days 6–20): Formal application to distribution utility (REB / DESCO / DPDC / BPDB) for Net-Metering feasibility clearance.

3. Stage 3: SREDA Official Registration & NOC (Days 21–35): Submission of technical dossiers to Sustainable and Renewable Energy Development Authority for gazetted approval.

4. Stage 4: High-Spec Procurement & Installation (Days 36–75): Fast-track customs clearance of Tier-1 equipment, mechanical mounting, and switchgear cabling.

5. Stage 5: Joint Utility Inspection, Bi-Directional Meter Sync & COD (Days 76–90): Witness testing with utility inspectors, calibration of Class 0.2S meter, and official Commercial Operation Date commissioning.

7. 25-YEAR O&M; & PERFORMANCE GUARANTEE SLA

Maximizing Asset Yield, Robotic Panel Washing & 24/7 Cloud Telemetry from Ariel Command Hub

A solar power plant is a 25-year financial asset. Its lifetime financial yield depends entirely on proactive Operations and Maintenance (O&M;). Ariel Renewable Energy provides comprehensive post-commissioning service level agreements (SLAs) to guarantee maximum generation:

O&M; Service Pillar	Operational Execution Protocol	Contractual Guarantee
Performance Ratio (PR) Guarantee	Continuous automated comparison of actual kilowatt-hour generation against irradiance data gathered by on-site calibrated pyranometers. Weather-corrected PR audited quarterly.	**Guaranteed PR ≥ 80.0%** Contractual liquidated compensation for shortfalls.
Quarterly Drone Thermography	Autonomous aerial radiometric thermal imaging scans all solar panels to detect micro-cracks, module hot-spots, diode failures, and PID degradation before output drops.	**Full Thermographic Audit Report** delivered to client management quarterly.
Bi-Monthly Panel Washing Protocols	Scheduled panel cleaning using demineralized, low-TDS water and soft microfiber brushes during dawn/dusk hours to prevent thermal shock and remove dust, soot, and industrial exhaust film.	Maintains panel soiling loss strictly **< 2.5%** throughout the dry winter months.
24/7 Cloud Telemetry & Rapid Dispatch	Every inverter string is continuously monitored via Ariel-SaaS™ Cloud Telemetry. Any inverter trip, insulation drop, or grid surge triggers automated technician dispatch.	**4-Hour Emergency On-Site Response** in Gazipur, Narayanganj, Savar, Dhaka & Chattogram.



Figure 4: Ariel Renewable Energy Technology Center & Central Grid Synchronization Station coordinating real-time national fleet dispatch.

8. PRELIMINARY SITE SURVEY & FEASIBILITY AUTHORIZATION

Non-Binding Executive Consent for Free 3D Drone Roof Mapping & Electrical Substation Audit

To provide your board with an exact, bankable investment proposal customized specifically to your facility's roof structure, connected load, and utility billing history, Ariel Renewable Energy conducts a comprehensive, non-intrusive on-site engineering survey. Execution of this authorization grants our engineering team permission to access the site.

SCOPE OF FREE PRELIMINARY FEASIBILITY AUDIT (ZERO COST TO CLIENT):

- **3D Drone Aerial Photogrammetry:** Precision high-definition drone roof scan to map usable area, skylights, vents, and shadow obstacles.
- **Structural Roof Load Verification:** Visual and mechanical inspection of PEB purlins, rafters, sheet gauges, or RCC slab rebar.
- **Electrical Substation Audit:** Assessment of transformer rating (kVA), LT/HT switchgear, busbar capacity, and Net-Meter integration point.
- **Historical Utility Bill Analysis:** Evaluation of past 12 months' electricity bills to model peak demand and net-metering cashflow yield.
- **Deliverable within 5 Working Days:** A bespoke **4-Page Bankable Roof Feasibility Dossier** containing PVsyst 3D solar layout, guaranteed monthly generation tables, exact financial ROI, and debt financing advisory.

AUTHORIZED ON BEHALF OF CLIENT FACILITY:	ACCEPTED BY ARIEL RENEWABLE ENERGY LIMITED:
_____ **Authorized Signatory Signature**	_____ **Ariel Energy Project Director Signature**
Full Name: _____	**Full Name:** Syed Akramul Hassan
Designation: Chairman / MD / Director / CEO	**Designation:** Managing Director & CEO
Company / Facility Name: _____	**Corporate Entity:** Ariel Renewable Energy Limited
Factory / Site Address: _____	**Parent Group:** ArielMC Enterprise Eco-System
Contact Mobile / Email: _____	**Command Hub:** energy.arielmc.com | crm.arielmc.com
Date: ____ / ____ / 2026	**Date:** ____ / ____ / 2026
Corporate Seal / Stamp:	**Official Corporate Seal:**